The Polynomial Certificate of Transputation: MDL Unification, Quantum Measurement, and the SRRG Fixed Point

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Abstract

The quantum measurement problem — why a superposition produces a single definite outcome, and why with probability $|c_k|^2$ — remains unresolved across standard interpretations of quantum mechanics, which either postulate collapse, multiply unobservable branches, or explain apparent classicality without resolving outcomes. This paper provides a formal derivation of the measurement mechanism from first principles within the GTE (Generative Triple Evolution) framework.

The GTE polynomial $p(L, C, R) = C + R - CR - LCR$ over GF(7), the unique Minimum Description Length (MDL)-minimal encoding of the Standard Model generation structure, carries a complete formal certificate of the quantum measurement mechanism. Its Perfect Self-Containment (PSC) projection is f_{MDL}, the discrete adjudicator that defines the measurement orbit $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$ — the only PSC-consistent trajectory in the cellular automaton dynamics — and this projection is characterized in three parts: f_{MDL} agrees with p on the binary sector $\{0, 1\}^3$ (Rule 110), overrides p on the ten SM-orbit neighborhoods to enforce generation-orbit consistency, and projects to zero on all remaining 325 neighborhoods. The polynomial's real diagonal encodes the Self-Referential Renormalization Group (SRRG) adjudication fixed point $\varphi = (\sqrt{5} - 1)/2 \approx 0.618$ (machine-certified, $\mathbf{A}_{\mathrm{Lean}}$). A Three-Level MDL Unification Theorem (**A/D**) shows that $P^{\top}(X) = \arg\min_{x \in X} K(x \mid \mathrm{PSC})$ operates at three nested, non-circular domains: theory selection (p from rule space), field dynamics (Φ_{MDL} via the PMDL variational principle), and quantum event adjudication (Born rule via transputation). The polynomial p is the connecting object: it *is* the output of Level 1, enters the PMDL action at Level 2, and its PSC-projection f_{MDL} defines the orbit used at Level 3. Differential Self-Adjudicative Computation (DSAC) is positioned as the continuum member of the $[D]$ adjudicator class, bridging the discrete f_{MDL} projection and the non-computable transputation selector $P_D^{\top}$.

Claim-Strength Taxonomy

Every result in this paper carries one of five certification levels:

- $\mathbf{A}_{\text{Lean}}$ (**A-Lean**) — Lean 4 machine proof, zero `sorry`, zero custom axioms. Conditional $\mathbf{A}_{\text{Lean}}$ uses a named physics axiom (always stated explicitly).
- **A/D** (**A/D**) — Analytic or fully decidable proof; human-verifiable.
- **A** (**A**) — Computationally verified; reproducible from published scripts.
- **B** (**B**) — Theoretically motivated conjecture with numerical support.
- **D** (**D**) — Speculative hypothesis; further development needed.

Ordering: $\mathbf{A}_{\text{Lean}} > \mathbf{A/D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$. All Lean certs use `ugp-lean` (canonical public repo) unless stated otherwise; module paths are given in Appendix A.

1 Introduction

The measurement problem — how quantum superposition becomes a definite outcome — remains unresolved in most interpretations of quantum mechanics. Standard frameworks postulate measurement as an axiom (Copenhagen [1]), multiply branches without a selection mechanism (Everett [4]), rely on decoherence without resolving outcomes [24], or postulate physical collapse [5]. CA-based interpretations of QM also exist [21,23]. This paper provides a formal derivation of the measurement mechanism from first principles within the GTE framework: quantum measurement is not added as an axiom but is forced — by PSC closure and diagonal self-reference — as a consequence of the universe being a record-bearing, reflexive computational system [10,17].

The GTE polynomial $p(L,C,R) = C + R - CR - LCR$ over GF(7) is, from first principles, the unique MDL-minimal CA rule consistent with the Standard Model generation structure and PSC [14,16]. Three prior papers establish parts of the measurement story: P49 [16] proves the four-object tower $p \to f_{\text{MDL}} \to \text{CMCA} \to \Phi_{\text{MDL}}$ and the invariant-subset uniqueness of f_{MDL}; P37 [17] derives the full Hilbert space from the f_{MDL} ring and the Born rule from information-theoretic loss; and P48 [10] establishes the transputation framework (D_1–D_5 constraints, forcing theorems) and the MDL three-roles non-circularity argument (P48 §5, "The Three Faces of MDL").

The present paper establishes what none of those papers state explicitly:

(i) The **three-part characterization of f_{MDL} as PSC-correction of p** — binary agreement (Rule 110), SM-orbit override, vacuum zeroing — with the information-theoretic gap between p and f_{MDL} computed at both the distributional level (KL divergence: 4.6–7.7 bits) and the algorithmic level (K-gap: 31 bits).

(ii) The **SRRG fixed-point bridge**: $p_{\mathbb{R}}(x,x,x) = x$ has the positive real root $\varphi = (\sqrt{5}-1)/2$, identically equal to the SRRG adjudication fixed point — machine-certified ($\mathbf{A}_{\text{Lean}}$, `gte_poly_srrg_bridge`).

(iii) The **Three-Level MDL Unification Theorem** (**A/D**): $P^{\top}(X) = \arg\min K(x \mid \text{PSC})$ operates with p as the connecting object across all three levels.

(iv) The **complete discrete measurement mechanism**: from p-dynamics (Class 3 chaotic attractor of period 475) through the PSC-projection π_{PSC} to the f_{MDL} orbit, to transputation as the non-computable continuum realization.

(v) **DSAC as the $[D]$-member** bridging the discrete f_{MDL} projection and the non-computable $P_D^{\top}$.

Section 8 locates the GTE transputation mechanism relative to Copenhagen, many-worlds, Bohmian, GRW, and decoherence approaches, explaining why transputation constitutes a structurally distinct fourth option [10]. Section 2.1 (within Section 2) connects the f_{MDL} certificate to Φ_{MDL} via the Algebraic Lifting Theorem, identifying the continuum significance of the three-part characterization.

Reading order. This paper builds on P49 (Objects 0–T, f_{MDL} structure), P37 (Hilbert space and Born rule from the f_{MDL} ring), P48 (transputation framework, MDL three roles, D_1–D_5 constraints), and P13 [15] (DSAC architecture, TE2.U). Each of those papers is cited rather than reproduced here.

2 The GTE Computational Hierarchy

The GTE framework has a four-level architecture in which each level is the continuum extension or PSC-projection of the level below:

$$\underbrace{p}_{\text{Object 0}} \xrightarrow{\pi_{\mathrm{PSC}}} \underbrace{f_{\mathrm{MDL}}}_{\text{Object 1}} \xrightarrow{\text{CMCA embedding}} \underbrace{\text{CMCA tape}}_{\text{Object 2}} \xrightarrow{\text{Algebraic Lifting}} \underbrace{\Phi_{\mathrm{MDL}}}_{\text{Object 3/T}} . \tag{1}$$

Object 0 is the MDL-minimal polynomial; Object 1 is its PSC-restriction to the admissible orbit; Object 2 embeds Object 1 into a 1+1D chiral CA tape with Rule 110/Rule 124 pair and an asynchronous fractal cellular automaton (AFCA) timing clock; Object T is the continuum field limit. These objects are pairwise distinct: p is a degree-3 polynomial, f_{MDL} is provably non-polynomial (Schwartz–Zippel, **A**), the CMCA tape is a dynamical system in 1+1D, and Φ_{MDL} is a $\mathbb{Z}_7$-symmetric Klein–Gordon scalar field [16, 19, 20].

The present paper is concerned with the transition $p \xrightarrow{\pi_{\mathrm{PSC}}} f_{\mathrm{MDL}}$ and its extension through the adjudicator class $[D]$ to the quantum measurement selector $P_D^{\top}$. The transputation layer $[D]$ sits *above* Object T in the adjudication hierarchy:

$$p \;\to\; f_{\mathrm{MDL}} \;\to\; \mathrm{DSAC} \;\in\; [D] \;\to\; P_D^{\top} \text{ (non-computable)}. \tag{2}$$

The CMCA and Φ_{MDL} levels are the *physical substrate*; the $[D]$-layer is the *adjudication mechanism* operating on top of that substrate. For the full architecture, see P48 [10].

2.1 Connection to Φ_{MDL} via Lifting and Descent

The Algebraic Lifting Theorem ($\mathbf{A}_{\mathrm{Lean}}$, P43 [11]) maps the CMCA algebraic structure (Level 2) to Φ_{MDL} (Level 3) exactly; the Algebraic Descent Theorem ($\mathbf{A}_{\mathrm{Lean}}$, P48 [10]) confirms that the same algebraic structure persists at every finite resolution. The discrete measurement certificate established in this paper therefore has a direct continuum interpretation. The three-part f_{MDL} characterization — binary agreement, orbit override, vacuum zeroing — lifts to a field-theoretic statement: the PSC-admissible kink sector of Φ_{MDL} carries the same three-part structure at continuum scale. The discrete transputation event $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$ lifts to a Φ_{MDL} field-configuration transition: a BPS kink evolving through its three-generation cascade before settling to vacuum. The non-computability of Level $2 \to 3$ lifts to the statement that the Φ_{MDL} path integral's branch selection is non-computable — the formal content of the D_3 constraint.

3 f_{MDL} as PSC-Correction of p

3.1 Three-Part Characterization

Objects 0 and 1 are not the same function. Their relationship is a three-part structure:

Three-Part Characterization of f_{MDL}

Part A — Binary sector $\{0,1\}^3$, 8 inputs ($\mathbf{A}_{\mathrm{Lean}}$):
$f_{\mathrm{MDL}}(L,C,R) = p(L,C,R)$ for all $(L,C,R) \in \{0,1\}^3$. Both give the Rule 110 truth table [2]. Machine-certified: `p_poly_agrees_fmdl_on_binary` (`UgpLean/GTE/Z7InvariantSubsets.lean`).

Part B — SM orbit neighborhoods, 10 inputs ($\mathbf{A}_{\mathrm{Lean}}$):
f_{MDL} overrides p with PSC-forced orbit values. Witness: $p(1,1,5) = 3$ but $f_{\mathrm{MDL}}(1,1,5) = 2$ (enforces $\mathrm{GEN}_1 \to \mathrm{GEN}_2$ winding transition). Machine-certified: `p_fmdl_disagree_on_orbit` (`UgpLean/Polynomial/PolyExplorations.lean`).

Part C — Remaining 325 neighborhoods ($\mathbf{A}_{\mathrm{Lean}}$):
$f_{\mathrm{MDL}}(L,C,R) = 0$ (vacuum projection). Machine-certified: `fmdl_zero_on_free_neighborhoods` (`UgpLean/Universality/CUP3DUniqueness.lean`).

Physical meaning of Part B (orbit override). The ten SM orbit neighborhoods where $f_{\mathrm{MDL}} \neq p$ are precisely the inputs that carry the Standard Model's generational structure. The raw polynomial does not encode this structure: at $(1,1,5)$, for example, p outputs 3 where the SM generation cascade requires 2. The f_{MDL} correction at these inputs is the PSC content: the CA learns to output what the Standard Model's generation cascade requires, not what the raw polynomial gives. This correction is not arbitrary — it is uniquely forced by the PSC orbit constraint; no other correction produces the $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$ orbit.

The three parts combine into a single certified statement:

Theorem 3.1 ($\mathbf{A}_{\mathrm{Lean}}$ — `psc_projection_gives_fmdl`). *f_{MDL} is the PSC-projection of p: it equals p on the binary ether sector, equals the PSC-forced orbit values on the ten SM-orbit neighborhoods, and equals zero on all other inputs. Equivalently, for any $(L,C,R) \notin \{0,1\}^3 \cup \{\mathit{SM\ orbit\ nbhds}\}$: $f_{\mathrm{MDL}}(L,C,R) = 0$.*

Proof (machine-assisted). The bundle `psc_projection_gives_fmdl` assembles `fmdl_zero_on_free_neighborhoods` (Part C; all $7^3 = 343$ inputs verified by `decide`) and `p_poly_agrees_fmdl_on_binary` (Part A). Part B follows from `p_fmdl_disagree_on_orbit` as a witness. Module paths are listed in Appendix A. □

Remark 3.2. f_{MDL} is *not* p restricted to a subset — it is p *corrected* on the SM orbit to enforce PSC consistency, then zero elsewhere. The correction encodes the PSC content: f_{MDL} is what p "should have been" to satisfy the physical generation-orbit constraint.

3.2 Information-Theoretic Gap

The PSC-projection $p \to f_{\mathrm{MDL}}$ has two distinct information-theoretic signatures.

Algorithmic complexity gap ($\mathbf{A}_{\mathrm{Lean}}$, A). The MDL description of p requires 19 bits (`mdl_ca_rule_coding_closed`, $\mathbf{A}_{\mathrm{Lean}}$, P49) [16]. The MDL description of f_{MDL} requires approximately 50 bits (the 8-entry Rule 110 table plus the 10 orbit-override entries, each with position and value).

The algorithmic K-gap is $K(f_{\mathrm{MDL}}) - K(p) \approx 31$ bits [7, 8]: the PSC orbit specification is more expensive to encode than the polynomial itself.

Distributional KL divergence (A). Treating p and f_{MDL} as probability distributions over $\mathrm{GF}(7)^3$ (normalized by the sum of positive outputs), the KL divergence is:

$$\mathrm{KL}(\tilde{f}_{\mathrm{MDL}} \parallel \tilde{p}) \;\approx\; 4.6\text{–}7.7 \text{ bits}, \tag{3}$$

depending on the model for the SM-orbit entries (Model A: binary only; Model B: binary plus orbit windows). The reverse divergence $\mathrm{KL}(\tilde{p} \parallel \tilde{f}_{\mathrm{MDL}}) \approx 24$–$25$ bits reflects the much higher entropy of p as a distribution.

The KL divergence and K-gap are not proportional: KL measures distributional discrepancy at the Shannon level [3, 9] (how different the output histograms are over $\mathrm{GF}(7)^3$), while the K-gap measures the structural overhead of specifying *which* inputs are PSC-admissible. Both quantify aspects of the $p \to f_{\mathrm{MDL}}$ gap, but at fundamentally different levels of description.

The distributional non-equivalence is machine-certified:

Theorem 3.3 (**A**$_{\mathrm{Lean}}$ — `kl_divergence_fmdl_p_nonzero`)**.** *There exists a triple* $(L, C, R) \in \mathrm{GF}(7)^3$ *such that* $p(L, C, R) \neq f_{\mathrm{MDL}}(L, C, R)$*, certifying that* p *and* f_{MDL} *are distinct as distributions over* $\mathrm{GF}(7)^3$.

3.3 The Attractor Structure of p

Under iteration of p on the $\mathbb{Z}_7^5$ ring (the 5-cell periodic ring encoding the SM generation winding triples), the dynamics have a precise bifurcation:

Theorem 3.4 (**A**$_{\mathrm{Lean}}$ — `period_475_returns`, `period_475_is_minimal`)**.** *The map* p *acting on* $\mathrm{GF}(7)^5$ *has an attractor of minimal period exactly 475. The vacuum basin contains 52 states (0.31% of the 16,807-state space); 99.69% of states asymptote to the period-475 attractor.*

In contrast, under f_{MDL}, the SM generation orbit $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$ is the *unique* attractor:

Theorem 3.5 (**A**$_{\mathrm{Lean}}$ — `fmdl_z7_three_generation_orbit`)**.** *The restriction of* f_{MDL} *to the 5-cell ring produces the three-generation orbit* $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$*, encoding the* $\mathbb{Z}_7$ *winding sequence of the Standard Model fermion generations.*

Theorem 3.6 (**A**$_{\mathrm{Lean}}$ — `fmdl_gen1_is_garden_of_eden`)**.** $\mathrm{GEN}_1 = [1, 5, 2, 2, 1]$ *is a Garden of Eden under* f_{MDL}*: it has no predecessor under the* f_{MDL} *rule. This makes* GEN_1 *the unique "first cause" of the physical orbit and explains why it is the lightest (most arithmetically stable) SM generation.*

The contrast between p and f_{MDL} is therefore stark: p exhibits Class 3 chaotic dynamics with a period-475 attractor; f_{MDL} has a unique, directed, 3-step orbit with no predecessors at its start. The PSC-projection selects the only trajectory that is physically realizable.

4 The SRRG Fixed-Point Bridge

The GTE polynomial over $\mathbb{R}$ (extending p from $\mathrm{GF}(7)$ to $\mathbb{R}$ by the natural embedding) yields the real-valued function $p_{\mathbb{R}}(L, C, R) = C + R - CR - LCR$. Its diagonal restriction is:

$$p_{\mathbb{R}}(x, x, x) = 2x - x^2 - x^3. \tag{4}$$

The fixed-point equation $p_{\mathbb{R}}(x,x,x) = x$ reduces to:

$$x^3 + x^2 - x = 0 \implies x(x^2 + x - 1) = 0, \tag{5}$$

with roots $x = 0$, $x = (\sqrt{5}-1)/2 \approx 0.618$, and $x = -(1+\sqrt{5})/2 \approx -1.618$. The unique positive real root is

$$\varphi \;=\; \frac{\sqrt{5}-1}{2} \;\approx\; 0.618034. \tag{6}$$

This value is identically the *SRRG adjudication fixed point*: the Self-Referential Renormalization Group (P27 [18]) identifies the self-consistent running-coupling fixed point $g^* = (\sqrt{5}-1)/2$ (equivalently $1/\varphi_{\text{large}}$ with $\varphi_{\text{large}} = (1+\sqrt{5})/2 \approx 1.618$).

Theorem 4.1 (**A**$_{\text{Lean}}$ — `gte_poly_srrg_bridge`)**.** *The unique positive real fixed point of $p_{\mathbb{R}}(x,x,x) = x$ is $\varphi = (\sqrt{5}-1)/2$, identically equal to the SRRG adjudication fixed point g^*. Machine-certified in `srrg-lean` (`SRRGCABridge.lean`, zero sorry).*

Remark 4.2 (Physical significance of Theorem 4.1)*.* The polynomial p — derived from MDL minimality and PSC without reference to continuous dynamics — already encodes the adjudication attractor of the renormalization flow. The real diagonal of p does not merely "accidentally" produce φ: φ is the unique value at which MDL-minimal theory selection (Level 1 of the hierarchy) and the continuum running-coupling fixed point (Level 2) agree. This is the first of two places where the polynomial bridges levels; the second is the Born-rule connection established in Section 5.

5 Three-Level MDL Unification Theorem

5.1 Statement

The MDL three-roles framework is established in P48 [10] and machine-certified as `mdl_tower_bundle` and `mdl_tower_three_levels_non_circular` (`MDLTower.lean`, ugp-lean, **A**$_{\text{Lean}}$). The present theorem is distinct: it identifies the *polynomial p itself as the connecting object* that unifies all three levels through a single arithmetic formula.

Theorem 5.1 (Three-Level MDL Unification, **A/D**)**.** *The formula*

$$P^{\top}(X) \;=\; \arg\min_{x \in X} \; K\big(x \mid \textit{PSC constraints}\big) \tag{7}$$

operates at three nested, logically non-circular levels, with p as the connecting object:

Level	*Transition*	*Domain X*	*Result*	*Cert*
$0 \to 1$	*Theory selection*	*All $\mathbb{Z}_N$ CA rules*	*p (19-bit polynomial, MDL-unique)*	$\boldsymbol{A}_{\text{Lean}}$
$1 \to 2$	*Field dynamics*	*Φ_{MDL} field configurations*	*PMDL variational solution; $\nabla^2\Phi_{\text{MDL}} = G_{\text{eff}}\, p(w_x, w_y, w_z)$*	***A/D***
$2 \to 3$	*Event adjudication*	*Realizations consistent with record w*	*Born rule $P(k) = \lvert c_k\rvert^2$ via $P_D^{\top}$*	$\boldsymbol{A}_{\text{Lean}}$

The polynomial p connects all three levels: p is the output of Level $0 \to 1$; p enters the PMDL action at Level $1 \to 2$ as the coupling functional; and p's PSC-projection f_{MDL} defines the orbit used by $P_D^{\top}$ at Level $2 \to 3$.

Proof sketch (CatAD). Level $0 \to 1$ ($\boldsymbol{A}_{\mathrm{Lean}}$). The MDL uniqueness chain proves that $K_{\mathrm{CA}} = 19$ bits is the minimum over all $\mathbb{Z}_N \times \mathbb{Z}_M$ CA rules compatible with PSC; the minimizer is p uniquely (`mdl_ca_rule_coding_closed`, `t96_02_mdl_selects_rule110`; see P49 T96-02 [16]).

Level $1 \to 2$ ($\boldsymbol{A/D}$). The PMDL action $S_{\mathrm{PMDL}} = \int \Phi_{\mathrm{MDL}} \cdot p(w_x, w_y, w_z)\, d^3x$ is the unique MDL-minimal action with $\mathbb{Z}_7$ symmetry and the CMCA algebraic certificate; its variational equation yields $\nabla^2 \Phi_{\mathrm{MDL}} = G_{\mathrm{eff}} \cdot p(w_x, w_y, w_z)$ (`pmdl_variational_gives_poisson`, `unique_cubic_gravity_coupling`, $\mathbf{A}_{\mathrm{Lean}}$; see P44–P47).

Level $2 \to 3$ ($\boldsymbol{A}_{\mathrm{Lean}}$). PSC closure + diagonal self-reference force $P_D^{\top}$ as the internal adjudicator (`closed_choice_forces_transputation`, transputation-lean, zero custom axioms [10]). The D_5 constraint forces Born-rule consistency: $P(k) = |c_k|^2$ (`born_rule_unconditional`, `UgpLean/Universality/PSCEffectMeasure.lean`, $\mathbf{A}_{\mathrm{Lean}}$). The realizations are defined over the f_{MDL} orbit, connecting this level to the Level $0 \to 1$ output.

Non-circularity. The three domains are strictly nested: theory space $\supset$ field configurations $\supset$ candidate realizations. No application of (7) at one level refers to any object at another level. The non-circularity is machine-certified: `mdl_tower_three_levels_non_circular` (`MDLTower.lean`, ugp-lean, $\mathbf{A}_{\mathrm{Lean}}$; see P48 [10]). $\square$

5.2 Bundle Certificate

Bundle Certificate ($\mathbf{A}_{\mathrm{Lean}}$, zero sorry): The cross-module bundle `mdl_three_level_unification` (`Polynomial/MDLThreeLevelUnification.lean`, ugp-lean) certifies the Three-Level Unification at the machine-verified level, wrapping:

- Level $0 \to 1$: `mdl_ca_rule_coding_closed` ($\mathbf{A}_{\mathrm{Lean}}$, `MDLDerivabilityCriterion.lean`);
- Level $1 \to 2$: `gte_poly_srrg_bridge` ($\mathbf{A}_{\mathrm{Lean}}$, `SRRGCABridge.lean`);
- PSC link $p \to f_{\mathrm{MDL}}$: `psc_projection_gives_fmdl` ($\mathbf{A}_{\mathrm{Lean}}$, `PolyExplorations.lean`);
- Level $2 \to 3$ measurement-orbit uniqueness — `fmdl_orbit_is_unique_psc_trajectory` ($\mathbf{A}_{\mathrm{Lean}}$; `CUP3DUniqueness.lean`).

Level $2 \to 3$ adjudication forcing under PSC and record-divergent choice is certified separately by `closed_choice_forces_transputation` ($\mathbf{A}_{\mathrm{Lean}}$; transputation-lean, via `mdl_level23_closed_choice_forces_transputation` in the same module).

5.3 The Computability Gap

The three levels differ in their computational character:

Proposition 5.2 (A/D). *Level $0 \to 1$ (selecting p from rule space) is* computable*: the PSC-restriction π_{PSC} on the finite space of $\mathbb{Z}_7$-state CA rules is an effective decision procedure. Level $2 \to 3$ (transputation $P_D^{\top}$) is* non-computable*: the D_3 constraint forces a diagonal barrier that no total-effective algorithm can resolve on the self-referential fragment of the record.*

Proof. Computability of Level $0 \to 1$: the space of 7-state 3-input CA rules has $7^{7^3} = 7^{343}$ elements; PSC filtering is an explicit finite enumeration; the MDL minimum is attained at p. Non-computability of Level $2 \to 3$: `asr_rt_not_computable` certifies that no total computable function can decide record-truth on the Arithmetic Self-Reference fragment (`UgpLean/SelfReference/`, $\mathbf{A}_{\mathrm{Lean}}$; P48 [10]). $\square$

Remark 5.3 (Structure of the computability gap). The level-raising map $\pi^{(0)}_{\mathrm{PSC}} \to \pi^{(\infty)}_{\mathrm{PSC}}$ is a *structural correspondence*, not a categorical functor. A functor would require morphism-preservation, but the computable morphisms at Level $0 \to 1$ map to non-computable morphisms at Level $2 \to 3$; no structure-preserving map exists across the computability boundary. The correspondence is therefore at the level of MDL semantics: both levels implement $\arg\min K(x \mid \mathrm{PSC})$, and non-computability emerges in the continuum limit.

The SRRG fixed point φ provides a *numerical bridge* across this computability gap: it is simultaneously the real-diagonal fixed point of p at Level 0 (Theorem 4.1, $\mathbf{A}_{\mathrm{Lean}}$) and the self-consistent adjudication attractor of the SRRG at Level 2 (P27 [18], $\mathbf{A}_{\mathrm{Lean}}$). The polynomial carries the adjudication answer in its algebraic structure, even though computing that answer in the full quantum case requires non-computable adjudication.

Remark 5.4 (Significance of Three-Level Unification). The significance of the Three-Level MDL Unification is not merely structural elegance. The same MDL principle that selects the theory from an astronomically large rule space also governs how that theory evolves as a field and how it adjudicates quantum events. These are not three separate applications of an information-theoretic principle — they are three domains in which a self-contained universe necessarily applies the same self-description criterion. The computability gap between Level $0 \to 1$ (computable PSC-restriction) and Level $2 \to 3$ (non-computable transputation) is not an inconsistency — it is the formal content of the Physical Incompleteness Theorem (analogous to Gödel's incompleteness [6] and Turing's undecidability [22]): a universe powerful enough to describe itself cannot compute all of its own event outcomes.

6 DSAC as $[D]$-Member and the Adjudication Hierarchy

6.1 The $[D]$ Adjudicator Class

The GTE-Möbius architecture $(A, e, [D])$ [13] designates $[D]$ as the equivalence class of all adjudication measures D satisfying the D_1–D_5 constraints:
D_1 Internal decidability: D operates from inside the record-bearing system.
D_2 PSC-consistency: D selects only PSC-admissible trajectories.
D_3 Non-computability: D is not a total-effective algorithm on the diagonal-capable fragment.
D_4 Unique minimum: D produces a single definite outcome.
D_5 Born consistency: outcome probabilities satisfy $P(k) = |c_k|^2$.
Any two measures satisfying D_1–D_5 are in the same PSC-coherence class $[D]$ (P48 [10]).

6.2 DSAC as a $[D]$-Member

Differential Self-Adjudicative Computation (DSAC, [12]) is explicitly identified in P48 as "one concrete member of $[D]$": DSAC's Δ-functional minimization satisfies D_1–D_5 by construction (NEMS Paper 77, **A/D**). The adjudication hierarchy is therefore:

$$p \xrightarrow{\pi^{(0)}_{\mathrm{PSC}}} f_{\mathrm{MDL}} \xrightarrow{\text{continuum PSC-min}} \mathrm{DSAC} \in [D] \xrightarrow{\text{non-comp. limit}} P_D^{\top}, \tag{8}$$

where:

- $p \to f_{\mathrm{MDL}}$: discrete, computable PSC-projection (Proposition 5.2).
- $f_{\mathrm{MDL}} \to \mathrm{DSAC}$: continuum extension (**B**; formal derivation open, Open Problem 1).

- DSAC $\to P_D^\top$: non-computability emerges from D_3 constraint.

Remark 6.1 (DSAC's role in the hierarchy). DSAC is the continuum PSC-minimizing dynamics (the PR-0 field substrate [15] with Δ-functional minimization). It occupies the intermediate position between the discrete, computable f_{MDL} projection and the continuous, non-computable $P_D^\top$ adjudication. The full chain (8) provides the complete picture of MDL-minimization at increasing physical scales.

7 The Discrete Measurement Mechanism

The preceding sections assemble into a complete formal account of discrete quantum measurement in the GTE framework:

Pre-measurement: Class 3 dynamics. The quantum state evolves under p-dynamics on $\mathrm{GF}(7)^5$. As shown in Theorems 3.1–4.1, 99.69% of initial states asymptote to the period-475 attractor; none of these trajectories pass through the PSC-admissible SM generation orbit. The state is a superposition over PSC-admissible p-trajectories.

Transputation fires: PSC-projection selects the orbit. The adjudicator $P_D^\top$ applies π_{PSC} — the MDL-minimal selection over all trajectories consistent with the macroscopic record w. By D_2, only PSC-consistent trajectories are selected; by D_4, a unique minimum is reached; by D_5, the probability of each outcome k is $|c_k|^2$.

Post-measurement: the f_{MDL} orbit. The post-measurement state is on the f_{MDL} orbit $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$, the unique PSC-admissible trajectory (`fmdl_z7_three_generation_orbit`, $\mathbf{A}_{\mathrm{Lean}}$). This *is* the realized physical sequence: the three SM generations decaying to vacuum. GEN_1 is a Garden of Eden (`fmdl_gen1_is_garden_of_eden`, $\mathbf{A}_{\mathrm{Lean}}$), so the orbit has no predecessor — it is the unique origination point of the physical sequence.

Theorem 7.1 ($\mathbf{A}_{\mathrm{Lean}}$ — `fmdl_orbit_is_unique_psc_trajectory`)**.** *Under deterministic f_{MDL} dynamics (`fmdl_step5`), the trajectory $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$ is the unique three-step PSC-admissible orbit from GEN_1: at each stage the successor state is uniquely determined, GEN_1 has no predecessor (Garden of Eden), and no alternative three-step PSC-consistent sequence exists. Machine-certified in `Universality/CUP3DUniqueness.lean`, zero `sorry`.*

Summary.

Theorem 7.2 (Discrete Measurement Mechanism, **A/D**)**.** *In the GTE framework, quantum measurement proceeds in three stages:*

(i) The pre-measurement quantum state is a superposition over p-compatible trajectories in $\mathrm{GF}(7)^5$.

(ii) Transputation $P_D^\top \in [D]$ applies π_{PSC}, selecting the unique PSC-admissible trajectory and producing a definite outcome with probability $|c_k|^2$.

(iii) The post-measurement state lies on the $\mathrm{GEN}_1 \to \mathrm{GEN}_2 \to \mathrm{GEN}_3 \to \mathrm{VAC}$ orbit, the unique fixed point of the f_{MDL} dynamics (Theorem 7.1, $\boldsymbol{A}_{\mathrm{Lean}}$).

This mechanism is non-algorithmic (Proposition 5.2), lawful (D_1–D_5 constraints), and fully internal to the PSC-closed system.

Remark 7.3. The Hilbert space structure, the spectral content, and the full Born rule derivation from information-loss on the f_{MDL} ring are established in P37 [17], which this theorem complements. P37 derives the measurement rule from the information-theoretic structure of the ring; the present paper identifies the *discrete CA mechanism* through which transputation selects the PSC-admissible trajectory. The two perspectives are mutually consistent and address complementary levels of description.

8 The Measurement Problem: GTE vs. Prior Frameworks

Quantum mechanics, as ordinarily formulated, requires an external rule — the Born rule together with wavefunction collapse — grafted onto unitary Schrödinger evolution. Unitary dynamics alone never selects a unique outcome; something outside the formalism must be invoked. This is the *measurement problem*.

Copenhagen / Bohr. The Copenhagen interpretation defines measurement by reference to a classical apparatus, without accounting for the quantum–classical boundary. In GTE, that boundary *is* the Level $0 \to 1$ transition ($p \to f_{\mathrm{MDL}}$): the "classical apparatus" corresponds to the PSC-admissible orbit filter.

Many-worlds / Everett. Everettian quantum mechanics [4] eliminates collapse: all branches exist, with the Born rule supplied by decision theory or branch-counting arguments rather than derivation. GTE's response: transputation is the unique PSC-forced branch selector. GTE does not branch — it selects exactly one PSC-consistent trajectory, non-computably but lawfully.

Bohmian mechanics / de Broglie–Bohm. Bohmian mechanics [1] assigns particles definite positions guided by a pilot wave; the theory is explicitly non-local. In GTE, the f_{MDL} orbit is the deterministic trajectory (deterministic CA dynamics); the pilot-wave role is played by the PSC-admissible sector of Φ_{MDL}. GTE is local (AFCA causal invariance); Bohm is not.

Objective collapse / GRW. GRW-type theories [5] postulate physical wavefunction collapse triggered by a stochastic process. In GTE, transputation is a physical collapse-like process, but forced by PSC (non-algorithmic, lawful) rather than postulated. The discrete mechanism ($p \to f_{\mathrm{MDL}}$ orbit) is GTE's derived collapse.

Decoherence approaches. Decoherence [24] explains apparent classicality and selects a preferred basis, but does not resolve the measurement problem: the Born rule remains an external postulate. In GTE, the Born rule is derived (four routes, $\mathbf{A}_{\mathrm{Lean}}$). Decoherence corresponds to the Level 0 chaotic dynamics of p; the transition to f_{MDL} dynamics is the decoherence event that resolves outcomes.

GTE's position. Transputation is the fourth option established in P48 [10]: (a) lawful (D_1–D_2), (b) non-algorithmic (D_3, forced by the Physical Incompleteness Theorem), (c) PSC-consistent (D_4), and (d) Born-consistent (D_5). It differs from collapse (no external rule), many-worlds (selects, not branches), Bohm (local, not guided), and GRW (derived, not postulated).

What Transputation Is Not

Transputation (the GTE measurement mechanism) is *not*:

- **Copenhagen collapse:** no external classical apparatus is invoked; the quantum-classical boundary IS the Level 0→1 transition ($p \to f_{\mathrm{MDL}}$).
- **Many-worlds branching:** exactly one PSC-consistent trajectory is realized, not all.
- **Bohmian pilot wave:** GTE is local (AFCA causal invariance); no non-local hidden variable.
- **GRW objective collapse:** transputation is derived from PSC, not postulated as an additional law.
- **Decoherence alone:** decoherence selects a preferred basis but does not resolve outcomes; transputation resolves outcomes.

Transputation IS: the unique PSC-consistent, non-computable, lawful internal adjudicator forced by the Physical Incompleteness Theorem (`closed_choice_forces_transputation`, $\mathbf{A}_{\mathrm{Lean}}$, `transputation-lean`).

9 Open Problems

Open Problem 1 (Continuum limit $f_{\mathrm{MDL}} \to$ DSAC)*.* Derive formally that DSAC is the continuum limit of the f_{MDL} PSC-projection, in the sense that the DSAC Δ-functional on Φ_{MDL} field configurations reduces to the discrete PSC-restriction $\pi^{(0)}_{\mathrm{PSC}}$ in the lattice limit. Current status: **B** (plausible from the PR-0 field substrate structure [15], no formal derivation).

Open Problem 2 (DSAC D_3 non-computability)*.* Prove that DSAC satisfies the D_3 constraint: DSAC cannot decide its own convergence on diagonal-capable input structures. This would upgrade DSAC's membership in $[D]$ from **A/D** (citation-based) to $\mathbf{A}_{\mathrm{Lean}}$. Lean candidate: `dsac_member_of_D_class`.

10 Conclusion

The central result of this paper is the Three-Level MDL Unification: a single MDL principle governs theory selection (p from rule space), field dynamics (Φ_{MDL} via PMDL), and quantum event adjudication (Born rule via transputation). The discrete measurement mechanism is established at **A/D**, with orbit uniqueness now machine-certified at $\mathbf{A}_{\mathrm{Lean}}$ (Theorem 7.1). The remaining gaps are well-posed mathematical questions — the continuum limit $f_{\mathrm{MDL}} \to$ DSAC (Open Problem 1) and DSAC D_3 certification (Open Problem 2) — not gaps in the framework. The cross-module Lean bundle for Three-Level Unification is now machine-certified (§5.2, `mdl_three_level_unification`).

This paper completes the polynomial certificate chain begun in P49 [16], situating the measurement mechanism relative to P37 [17] (Hilbert space and Born rule) and P48 [10] (transputation framework and MDL three roles). The GTE polynomial p is the connecting object: its PSC-projection f_{MDL} defines discrete measurement; its real diagonal encodes the SRRG fixed point; and it enters the PMDL action at continuum scale.

A Lean Certification Table

All Lean certifications cited in this paper. All theorems are in the canonical `ugp-lean` repository (zero sorry). Modules `Polynomial/PolyExplorations` and related were verified prior to publication.

Table 1: Lean 4 certifications cited in this paper. All entries are zero sorry. `UL` = ugp-lean; `SL` = srrg-lean; `TL` = transputation-lean.

Theorem name	Module	Repo	Cat
`psc_projection_gives_fmdl`	`Polynomial/PolyExplorations`	`UL`	$\mathbf{A}_{\text{Lean}}$
`fmdl_psc_projection_of_p`	`Polynomial/PolyExplorations`	`UL`	$\mathbf{A}_{\text{Lean}}$
`fmdl_zero_on_free_neighborhoods`	`Universality/CUP3DUniqueness`	`UL`	$\mathbf{A}_{\text{Lean}}$
`p_poly_agrees_fmdl_on_binary`	`GTE/Z7InvariantSubsets`	`UL`	$\mathbf{A}_{\text{Lean}}$
`p_fmdl_disagree_on_orbit`	`Polynomial/PolyExplorations`	`UL`	$\mathbf{A}_{\text{Lean}}$
`kl_divergence_fmdl_p_nonzero`	`Polynomial/PolyExplorations`	`UL`	$\mathbf{A}_{\text{Lean}}$
`fmdl_z7_three_generation_orbit`	`Universality/CUP3DUniqueness`	`UL`	$\mathbf{A}_{\text{Lean}}$
`fmdl_gen1_is_garden_of_eden`	`Universality/CUP3DUniqueness`	`UL`	$\mathbf{A}_{\text{Lean}}$
`fmdl_orbit_is_unique_psc_trajectory`	`Universality/CUP3DUniqueness`	`UL`	$\mathbf{A}_{\text{Lean}}$
`mdl_three_level_orbit_bundle`	`Universality/CUP3DUniqueness`	`UL`	$\mathbf{A}_{\text{Lean}}$
`mdl_three_level_polynomial_bundle`	`Universality/CUP3DUniqueness`	`UL`	$\mathbf{A}_{\text{Lean}}$
`mdl_three_level_unification`	`Polynomial/MDLThreeLevelUnification`	`UL`	$\mathbf{A}_{\text{Lean}}$
`mdl_level23_closed_choice_forces_transputation`	`Polynomial/MDLThreeLevelUnification`	`UL`	$\mathbf{A}_{\text{Lean}}$
`period_475_returns`	`Polynomial/PolyExplorations`	`UL`	$\mathbf{A}_{\text{Lean}}$
`period_475_is_minimal`	`Polynomial/PolyExplorations`	`UL`	$\mathbf{A}_{\text{Lean}}$
`gte_poly_srrg_bridge`	`Algebra/SRRGCABridge`	`SL`	$\mathbf{A}_{\text{Lean}}$
`srrg_equals_mdl_minimization`	`Algebra/SRRGCABridge`	`SL`	$\mathbf{A}_{\text{Lean}}$
`closed_choice_forces_transputation`	`Theorems/ForcedAdjudication`	`TL`	$\mathbf{A}_{\text{Lean}}$
`born_rule_unconditional`	`Universality/PSCEffectMeasure`	`UL`	$\mathbf{A}_{\text{Lean}}$
`mdl_tower_bundle`	`Framework/MDLTower`	`UL`	$\mathbf{A}_{\text{Lean}}$
`mdl_tower_three_levels_non_circular`	`Framework/MDLTower`	`UL`	$\mathbf{A}_{\text{Lean}}$

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Theorem name	Module	Repo	Cat
mdl_ca_rule_coding_closed	Classification/ MDLSelection	UL	$\mathbf{A}_{\text{Lean}}$
asr_rt_not_computable	SelfReference/	UL	$\mathbf{A}_{\text{Lean}}$

B Code and Reproducibility

The KL-divergence computation and structural analysis from which the numerical results in Section 3.2 are drawn are implemented in the scripts at:

- `papers/51_polynomial_transputation/scripts/p_transputation_kl_divergence.py` — Computes KL divergence, entropy, and information-theoretic gap between p and f_{MDL} as distributions over $\text{GF}(7)^3$; verifies the diagonal fixed-point φ.
- `papers/51_polynomial_transputation/scripts/p_transputation_formal_analysis.py` — Functor analysis, improved f_{MDL} model with orbit windows, and the Three-Level MDL correspondence.

Both scripts are self-contained, include wall-clock timeouts, and produce a JSON artifact `p_transputation_kl_results.json` with all numerical outputs. The scripts contain no local paths; all inputs are relative to the script directory.

To reproduce the distributional KL-divergence results from Section 3.2:

```
cd papers/51_polynomial_transputation/scripts
python3 p_transputation_kl_divergence.py
```

Expected output: KL(f_MDL || p) in the range $[4.6, 7.7]$ bits (Model A: 7.71 bits; Model B: 4.62 bits).

Lean verification. All theorems in Appendix A can be verified by building `ugp-lean` and `srrg-lean`:

```
cd /path/to/ugp-lean && lake build UgpLean
cd /path/to/srrg-lean && lake build
```

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